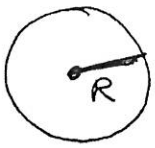
HW 07

P4.10



$$\vec{P}(\vec{r}) = k\vec{r}$$

(a) $\rho_b = -\vec{\nabla}\cdot\vec{P} = -k\,\vec{\nabla}\cdot\vec{r} = -3k$

as $\vec{\nabla}\cdot\vec{r} = \left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}\right)\cdot\left(x\hat{x} + y\hat{y} + z\hat{z}\right) = 3$

$$\sigma_b = \vec{P}\cdot\hat{r}\Big|_{r=R} = kR$$

(b) Electric field $\vec{E} = \vec{E}_\rho + \vec{E}_\sigma$

$\vec{E}_\rho$: field of a uniformly charged sphere, $\rho_b = -3k$

$\vec{E}_\sigma$: field of spherical shell $\sigma_b = kR$.

Inside the sphere: $r<R$

$\vec{E}_\sigma(r<R) = 0$ by Gauss's Theorem.

$$\int \vec{E}_\rho\cdot d\vec{S} = \frac{-3k\cdot\frac{4\pi}{3}r^3}{\varepsilon_0} \Rightarrow E_\rho\cdot 4\pi r^2 = -\frac{4\pi k}{\varepsilon_0}r^3$$

$$\vec{E}_\rho(r<R) = -\frac{k\vec{r}}{\varepsilon_0}$$

$$\vec{E}(r) = -\frac{k}{\varepsilon_0}\vec{r} \quad \text{for } r<R.$$

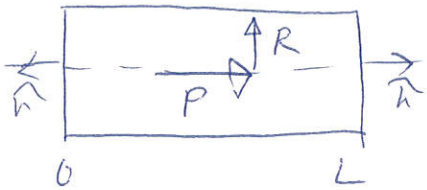
Outside the sphere: $r>R$

$$\left.\begin{aligned}\vec{E}_\sigma &= \frac{\sigma_b\cdot 4\pi R^2}{4\pi\varepsilon_0 r^2}\hat{r} = \frac{R^3 k\,\hat{r}}{\varepsilon_0 r^2}\\ \vec{E}_\rho &= \frac{\rho_b\cdot\frac{4\pi}{3}R^3}{4\pi\varepsilon_0 r^2}\hat{r} = -\frac{R^3 k\,\hat{r}}{\varepsilon_0 r^2}\end{aligned}\right\}\ \vec{E}(r) = 0 \text{ for } r>R.$$

Thus
$$\left.\begin{aligned}\vec{D}(r<R) &= \varepsilon_0\vec{E} + \vec{P} = -k\vec{r} + k\vec{r} = 0\\ \vec{D}(r>R) &= \varepsilon_0\vec{E} + \underbrace{\vec{P}}_{=0} = 0\end{aligned}\right\}\ D_r(r=R-0) = D_r(r=R+0)$$

as it should be.

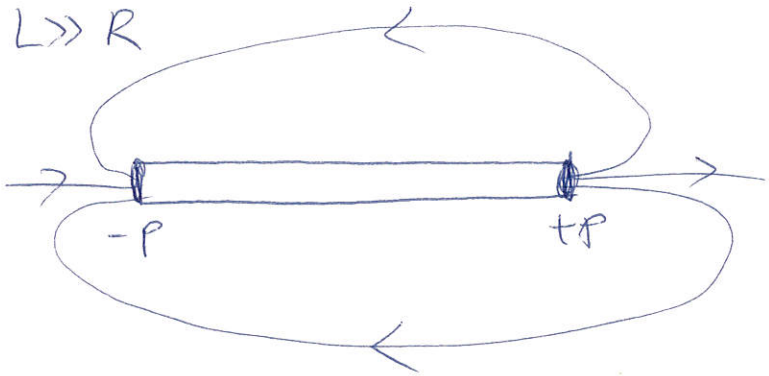
4.11



$\vec{P} = \text{const} \Rightarrow \rho_b = -\vec{\nabla}\cdot\vec{P} = 0$

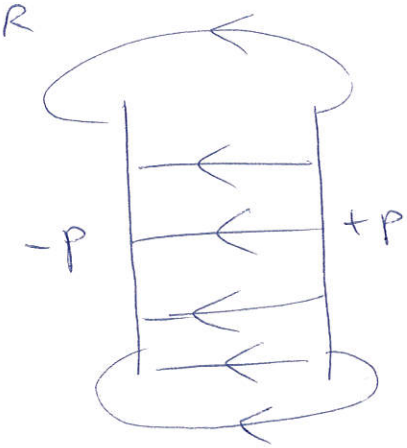
$$\sigma_b = \vec{P}\cdot\hat{n} = \begin{cases} +P, & x=L \\ -P, & x=0 \end{cases}$$

i) $L \gg R$



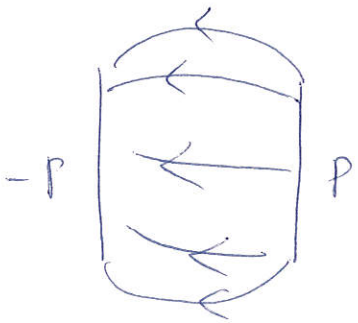
like two point charges

ii) $L \ll R$



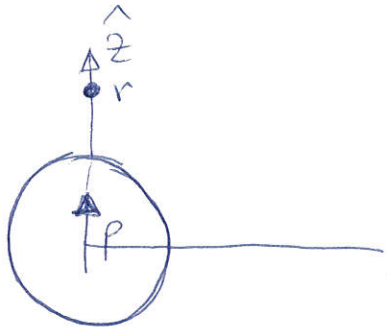
"Parallel-plate capacitor"

iii)



in-between

4.12



$$V(r) = \frac{1}{4\pi\varepsilon_0}\int_{r'\in \text{sphere}} \frac{(\vec{r}-\vec{r}\,')\cdot\vec{P}(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|^3}\, d^3\vec{r}\,'$$

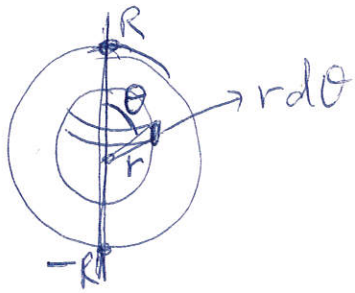
$$\vec{P} = P\hat{z}$$

let $\vec{r}$ be on z-axis so that $\vec{r} = \vec{z} = z\hat{z}$.

$\vec{r}\,'$ lives inside the sphere.

$$(\vec{r}-\vec{r}\,') = -x'\hat{x} - y'\hat{y} + (z-z')\hat{z}$$

$$V(r=z) = \frac{1}{4\pi\varepsilon_0}\int \frac{P(z-z')}{\left(x'^2+y'^2+(z-z')^2\right)^{3/2}}\, dx'dy'dz'$$



$$z' = r\cos\theta$$

$$\left.\begin{aligned} x' &= r\sin\theta\cos\varphi \\ y' &= r\sin\theta\sin\varphi \end{aligned}\right\} (x')^2+(y')^2 = r^2\sin^2\theta$$

$$\Rightarrow \quad x'^2+y'^2+(z-z')^2 = r^2\sin^2\theta + z^2 + r^2\cos^2\theta - 2zr\cos\theta = r^2+z^2-2rz\cos\theta$$

$$V(z) = \frac{P}{4\pi\varepsilon_0}\int \frac{z-r\cos\theta}{\left(z^2+r^2-2rz\cos\theta\right)^{3/2}}\, 2\pi r^2 dr \sin\theta d\theta$$

$$\int_0^\pi \sin\theta d\theta = -\int_1^{-1} d(+\cos\theta) = (u=\cos\theta) = \int_{-1}^{1} du$$

$$V = \frac{P\cdot 2\pi}{4\pi\varepsilon_0}\int_{-1}^{1} du \int \frac{z-ru}{\left(z^2+r^2-2rzu\right)^{3/2}}\, r^2 dr$$

$$I_1 = \int_{-1}^{1} \frac{du}{\left(z^2+r^2-2rzu\right)^{3/2}} = \int_{-1}^{1} d\left(\frac{1}{\left(z^2+r^2-2rzu\right)^{1/2}}\right) \frac{1}{rz}$$

$$= \frac{1}{rz} \left.\frac{1}{\sqrt{z^2+r^2-2rzu}}\right|_{-1}^{1} = \frac{1}{rz}\left(\frac{1}{\sqrt{z^2+r^2-2rz}} - \frac{1}{\sqrt{r^2+z^2+2rz}}\right) = \frac{1}{rz}\left(\frac{1}{|z-r|} - \frac{1}{|z+r|}\right)$$

$$I_2 = \int_{-1}^{1} du \frac{u}{\left(z^2+r^2-2rzu\right)^{3/2}} = \frac{1}{rz}\int d\left(\frac{1}{\sqrt{z^2+r^2-2rz}}\right) u$$

by parts $$= \frac{1}{rz}\left\{ \left.\frac{u}{\sqrt{z^2+r^2-2rzu}}\right|_{-1}^{1} - \int_{-1}^{1} \frac{du}{\sqrt{z^2+r^2-2rzu}} \right\}$$

$$= \frac{1}{rz}\left\{ \frac{1}{|z-r|} - \frac{-1}{|z+r|} + \frac{1}{rz}\int_{-1}^{1} \left(d\sqrt{z^2+r^2-2rzu}\right) \right\}$$

$$= \frac{1}{rz}\left\{ \frac{1}{|z-r|} + \frac{1}{|z+r|} + \frac{1}{rz}\left(|z-r| - |z+r|\right) \right\}$$

$$V = \frac{\rho}{2\varepsilon_0}\int_0^R dr\, r^2 \left(z I_1 - r I_2\right) = \frac{\rho}{2\varepsilon_0}\int_0^R dr\, r^2 \left\{ \left(\frac{1}{r} - \frac{1}{z}\right)\frac{1}{|z-r|} + \left(\frac{1}{r} + \frac{1}{z}\right)\frac{1}{|z+r|} - \frac{1}{rz^2}\left(|z-r| - |z+r|\right) \right\}$$

Because of $|z \pm r|$ factors, need to consider $z > r$ and $z < r$ separately

1) let $z > R$ first $\Rightarrow z > r$ for all $r \Rightarrow |z-r| = z-r$
$|z+r| = z+r$

$$V = \frac{P}{2\varepsilon_0}\int_0^R dr\, r^2 \left\{ \frac{z-r}{zr}\frac{1}{z-r} - \frac{z+r}{zr}\frac{1}{z+r} - \frac{1}{rz^2}\underbrace{(z-r-z-r)}_{-2r} \right\}$$

$$= \frac{P}{2\varepsilon_0}\int_0^R dr\, r^2 \left\{ \frac{2}{z^2} \right\} = \frac{P}{\varepsilon_0 z^2}\frac{R^3}{3} = \frac{\frac{4\pi}{3}PR^3}{4\pi\varepsilon_0 z^2} = \frac{P_{sph}}{4\pi\varepsilon_0 z^2}$$

where $\vec{P}_{sph} = \frac{4\pi}{3}R^3\vec{P}$ = is the total dipole moment of the sphere.

Note $P_{sph} = \vec{P}\cdot\hat{z}$

2) Now $z < R \Rightarrow$ the integral over (r) splits into sum

$$\int_0^R dr \ldots = \underbrace{\int_0^z dr \ldots}_{\text{here } r<z} + \underbrace{\int_z^R dr \ldots}_{\text{here } r>z \Rightarrow \begin{array}{l}|z-r| = r-z \\ |z+r| = r+z\end{array}}$$

$$V = \frac{P}{2\varepsilon_0}\int_0^z dr\, r^2 \left\{ \frac{2}{z^2} \right\} + \frac{P}{2\varepsilon_0}\int_z^R dr\, r^2 \left\{ \frac{z-r}{zr}\frac{1}{r-z} - \frac{z+r}{zr}\frac{1}{r+z} - \frac{1}{rz^2}\underbrace{(r-z-r-z)}_{-2z} \right\} =$$

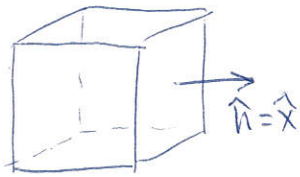
$$= \frac{P}{\varepsilon_0 z^2}\frac{z^3}{3} + \frac{P}{2\varepsilon_0}\int_z^R dr\, r^2 \underbrace{\left\{ \frac{-2}{rz} + \frac{2z}{rz^2} \right\}}_{=0}$$

$$= \frac{Pz}{3\varepsilon_0} = \frac{Pr\cos\theta}{3\varepsilon_0} \quad \text{because } z = r\cos\theta.$$

$\Rightarrow V(r) = \frac{Pr\cos\theta}{3\varepsilon_0}$ for r inside the sphere

coincides with Ex. 4.2

4.31



$$\vec{P} = k\vec{r}$$

$$\rho_b = -\vec{\nabla}\cdot\vec{P} = -3k$$

$\Rightarrow$ bulk bound charge

$$Q_{bulk} = \rho_b \cdot \underbrace{a^3}_{\text{volume}} = -3ka^3$$

Surface charge density

$$\sigma_b = \vec{P}\cdot\hat{n}$$

Consider right face where $\hat{n}=\hat{x}$

$$\sigma_b = k\vec{r}\cdot\hat{x} = kx = k\frac{a}{2}$$ because that face has $x=\frac{a}{2}$ for all points.

Total surface charge density

$$Q_{surf} = \sigma_b \cdot \underbrace{(\text{surface area})}_{6a^2} = k\frac{a}{2}\cdot 6a^2 = 3ka^3.$$

$\Rightarrow$ total bound charge density on the cube is 0.